Attaining the Entropy Limit for Union-Closed Families

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We prove that every finite union-closed family with a nonempty member has an element that lies in at least 0.38288525996… *of its members. This number is the exact limit of the entropy method in the form developed by Liu and Jiang, and earlier work had come within a hundred-millionth of it without reaching it. The proof chooses the auxiliary function so that the two inequalities behind the method touch zero at a single point, and shows by a convexity argument that they stay nonnegative around it. With a global coupling of Jiang, the bound is strict for every family. We also prove the conjecture itself for every family of at most fifty-two members, two more than was known, by reducing a smallest counterexample to two families of down-sets of an unknown order and refuting them with a checked satisfiability proof.*

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CONTENTS

I. Introduction 1
II. The entropy reduction 3
III. The construction at the limit 3
IV. Contact at one point and convexity around it 4
V. The rest of the unit square 5
VI. Proof of the theorems 6
VII. Beyond the limit 7
VIII. Families of at most fifty-two members 7
Data and code availability 9
Ethical declarations 9
Author contributions 9
A. Proof of the entropy reduction 9
B. Why the limit is C^* 9
C. The Hessians at the point of contact 10
D. The analytic estimates 10
E. The interval method 11
F. The strict inequality 11
G. The boundary configuration 11
References 12

I. INTRODUCTION

A family $\mathscr{F}$ of finite sets is *union-closed* if $A \cup B \in \mathscr{F}$ whenever $A, B \in \mathscr{F}$. Write $N = |\mathscr{F}|$ and, for an element i of the ground set,

$$p_i = \frac{|\{A \in \mathscr{F} : i \in A\}|}{N}, \qquad p = \max_i p_i.$$

Frankl's conjecture states that $p \geq \frac{1}{2}$ whenever $\mathscr{F}$ contains a nonempty set.[1] Let c_{UC} be the largest constant such that $p \geq c_{\mathrm{UC}}$ for every such family.

Until 2022 no positive constant was known. Gilmer [7] then proved $c_{\mathrm{UC}} \geq 0.01$ with a striking entropy argument: two independent uniform members of $\mathscr{F}$ have a union that is again a member, so its entropy is at most $\ln N$, while a short computation shows that the union has more entropy than either member when all frequencies are small. Within weeks Alweiss, Huang and Sellke [1], Chase and Lovett [4], Sawin [15] and Pebody [12] pushed the argument to $(3-\sqrt{5})/2 = 0.38196\ldots$. Sawin proposed to couple the two members instead of drawing them independently; Yu [17] and Cambie [3] developed this, and Liu [10] introduced conditionally independent couplings, proved 0.3823455… and found numerically that the method should reach 0.382709….[2]

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[1] The conjecture is usually dated to 1979 and attributed to P. Frankl; the survey of Bruhn and Schaudt [2] describes its history, its equivalent lattice and graph forms, and the many special classes for which it is known. The family $\{\emptyset, \{1\}\}$ shows that the constant $\frac{1}{2}$ cannot be raised, and the condition that some member is nonempty only excludes the family $\{\emptyset\}$. Every finite family on an arbitrary ground set can be reduced to one on a finite ground set with the same frequencies for the elements that matter, by keeping one element of a nonempty member and one element of each symmetric difference of two members.

[2] Liu's value 0.382709087918735 rested on numerically verified hypotheses. Costa [5] and Costa and Sadhu [6] proved it, and then, by varying the protocol of Liu's Example 5 and optimising two coefficients, proved

In September 2026 Jiang [9] gave a computer-assisted proof of $c_{\mathrm{UC}} \geq 0.38288525$. Jiang reduced the frequency bound to two scalar inequalities (K) and (M) on the unit square, and showed that no choice of the functions in them can give a constant above

$$C^* = \frac{1}{2} - \frac{(1-q^*)\ln(1+q^*)}{2\ln 2} = 0.3828852599667977965\ldots, \tag{1}$$

where $q^* = 0.6909077387254077545\ldots$ is the unique root in $(0,1)$ of

$$(1-q)\ln(1-q) + q\ln(1+q) = 0. \tag{2}$$

Costa and Sadhu [6] found the same limit for their inequality, and Moffat [11] proved that C^* bounds every single-letter certificate of this kind whose other protocols admit what Moffat calls component hiding.[3] Jiang notes that this theorem does not assert that C^* itself is achieved; the constant proved there lies 1.0×10^{-8} below it, and that of Costa and Sadhu 5.0×10^{-9} below it.

We show that the limit is attained.

Theorem A. *Every finite union-closed family that contains a nonempty set has an element lying in at least a fraction* $C^* = 0.3828852599667977965\ldots$ *of its members. In other words,* $c_{\mathrm{UC}} \geq C^*$.

Combining Theorem A with the global coupling of Jiang [9, Theorem 10.2] gives a strict inequality with an explicit margin for each family.

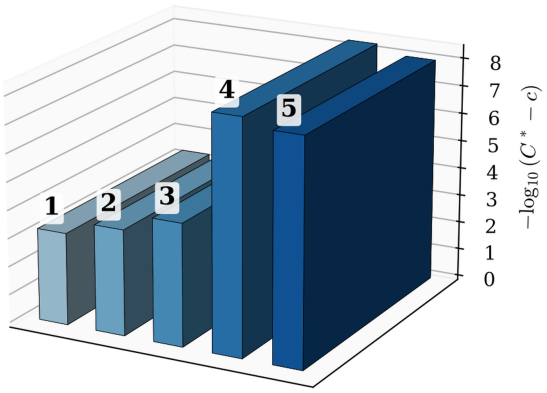
Theorem B. *If* $\mathcal{F}$ *is a union-closed family of size* $N \geq 2$ *that contains a nonempty set, then*

$$p \;\geq\; C^* + \frac{\ln^2(2N-1)}{128\,N^2\ln^2 N} \;>\; C^*.$$

The margin in Theorem B tends to zero, so it does not raise c_{UC} above C^*; by the results just quoted, going further needs an argument outside the single-letter framework. Figure 1 places the certified constants.

The second result of the paper concerns small families. Frankl's conjecture is known for every family with at most 50 members, which follows from the bound of Roberts and Simpson [13] on the size of a smallest counterexample and the verification of the conjecture for ground sets of at most 12 elements [2, 16]. We exclude the boundary case of that bound.

Theorem C. *Every union-closed family with at most* 52 *members that contains a nonempty set has an element lying in at least half of its members.*



1. Liu: 0.3823455000, gap 5.4e-04
2. Liu threshold: 0.3827090879, gap 1.8e-04
3. Moffat: 0.3828400000, gap 4.5e-05
4. Costa-Sadhu: 0.3828852550, gap 5.0e-09
5. Jiang: 0.3828852500, gap 1.0e-08

This paper: $c = C^* = 0.3828852599\ldots$, gap 0

FIG. 1. Certified lower bounds for c_{UC} in the entropy line of work, plotted as $-\log_{10}(C^* - c)$, so that a taller bar is closer to the limit C^*. The bound of this paper is C^* itself.

The proof (Section VIII) reduces a hypothetical counterexample with 51 members on 13 elements to two families of down-sets of an unknown partial order on 11 points, with four constraints derived by hand, and shows by a satisfiability computation with a checked proof certificate that no such pair exists. Without the computer, the same reduction proves that such a counterexample, of any size $4m-1$ on m elements, must contain two elements one of which lies in every member containing the other (Proposition VIII.3).

Why the limit was not reached before. At the limit there is no slack anywhere. The constant C^* comes from two necessary conditions: the edge $y = 1$ of (M) forces $mw_0 \geq \frac{1}{2}$, and the diagonal point $x = y = q^*$ of (M) together with $\psi(q^*)^2 \leq K(q^*, q^*) \leq \ln 2$ forces $1 - m \leq C^*$. To reach C^* all three inequalities must be equalities. Then (K) and (M) both vanish at the interior point (q^*, q^*), and an argument that certifies positivity box by box with a fixed margin cannot cover a neighbourhood of that point.[4]

What is new. We set every parameter to its limiting value: the profile is centred at q^* exactly, its scale is one, and $mw_0 = \frac{1}{2}$ exactly. We then prove that the two gaps vanish to second order at (q^*, q^*), with value and gradient zero (Proposition IV.1), and that their Hessians are positive definite on

$c_{\mathrm{UC}} \geq 0.3828852549667978$. Moffat [11] built a conditionally independent protocol whose kernel has the optimal diagonal and certified 0.38284 under two numerically checked hypotheses.

[3] Moffat's refined ceiling $c^{**} = 0.382885260\ldots$ coincides with C^*. The argument uses two adversarial laws of the conditional probabilities: a law that puts a tiny atom carrying all the entropy away from the atom where the element is surely absent forces the weight of the independent protocol to be at least $1/(2(1-c))$, and a diagonal two-point law then caps c. This ceiling is Moffat's result and we use it only to place ours.

[4] Jiang kept a margin of order 10^{-8} by scaling the profile with $\sigma = 0.99999999$ and taking $mw_0 = 0.500000008685975$; Costa and Sadhu kept a similar margin through a slightly smaller constant. With any positive margin the interval method closes, and the margin can be made as small as the computing budget allows, but it never vanishes.

the whole square $[q^*-0.008, q^*+0.008]^2$ (Proposition IV.2). A function with these two properties is nonnegative on the square. Outside it the inequalities are strict and the interval method of Jiang, rerun with the new parameters, covers the rest; three edge estimates that relied on the old margin turn out to hold with equality allowed (Section V). Figure 2 shows the two gaps over the unit square.

Plan. Section II states the entropy reduction. Section III fixes the functions at the limit and Section IV proves the contact at (q^*,q^*) and the convexity around it. Section V covers the rest of the square, Section VI proves the first two theorems and Section VII discusses what lies beyond C^*. Section VIII proves Theorem C. Appendices A to G give every derivation in full.[5]

II. THE ENTROPY REDUCTION

All logarithms are natural and $h(x) = -x\ln x - (1-x)\ln(1-x)$ is the binary entropy in nats, with $h(0)=h(1)=0$. Let $a:[0,1]\to[0,1]$ and put

$$r(x,y) = xy + a(x)a(y)\big(\min(x,y) - xy\big) \tag{3}$$

and $K(x,y) = h\big(r(x,y)\big)$. Let $\psi:[0,1]\to[0,\infty)$ and let $w_0, w_4 \ge 0$ with $w_0 + w_4 = 1$.[6]

Proposition II.1 (Jiang [9], Proposition 3.1)**.** *Suppose that $m>0$ and that for all $x,y\in[0,1]$*

$$K(x,y) \;\ge\; \psi(x)\psi(y), \tag{K}$$

$$2m\big[w_0 h(xy) + w_4\psi(x)\psi(y)\big] \;\ge\; xh(y) + yh(x). \tag{M}$$

Then every finite union-closed family containing a nonempty set has $p \ge 1-m$.

The proof, which we give in full in Appendix A, runs two sampling rules side by side. The first takes two independent uniform members. The second draws at each coordinate one shared uniform variable and lets each sample's next bit be zero with probability $x + a(x)(\mathbf{1}\{U\le x\} - x)$, where x is that sample's conditional probability of a zero; this keeps both samples uniform on $\mathscr{F}$ and makes the union bit zero with probability r. Inequality (K) bounds the entropy of the second union below by a product, conditional independence turns its expectation into a square, and (M) combines the two rules into the bound $w_0 H(Z_0) + w_4 H(Z_4) \ge \frac{1-p}{m}\ln N$. Both unions lie in $\mathscr{F}$, so the left side is at most $\ln N$.

The inequalities (K) and (M) cannot hold with $1-m > C^*$ [9, Proposition 4.1]; Appendix B repeats the short argument, because its equality cases are exactly the conditions that fix our construction.

[5] The authors' other work, on analytic estimates in number theory, is in [18, 19]; the archive of this paper is [21].

[6] The index 4 follows Jiang, who refers to Liu's Example 4; w_4 is a weight and not a power. The function a governs how strongly the two samples share randomness: $a\equiv 0$ gives independent bits and $a\equiv 1$ gives bits that are as positively correlated as their marginals allow.

III. THE CONSTRUCTION AT THE LIMIT

The construction has the shape of Jiang's, with every parameter at its limiting value. Put

$$f(q) = q\,h(q) - \tfrac12 h(q^2),$$

so that $f'(q) = -\big[(1-q)\ln(1-q) + q\ln(1+q)\big]$ and q^* is the unique maximiser of f on $(0,1)$.[7] Set

$$mw_0 = \tfrac12, \qquad mw_4 = \frac{f(q^*)}{\ln 2}, \qquad m = \tfrac12 + \frac{f(q^*)}{\ln 2} = 1 - C^*. \tag{4}$$

Thus $w_0 = 1/(2m) = 0.8102221287\ldots$ and $w_4 = 1-w_0$. The rational breakpoints are

$$A = \tfrac{179}{250}, \quad B = \tfrac{721}{1000}, \quad z = \tfrac{181}{256}, \quad L = \tfrac{1}{16}, \quad U = \tfrac{15}{16},$$

with $\frac12 < z < \frac{1}{\sqrt2} < A < B$.[8] Let

$$E(x) = -\frac{h'(q^*)}{2h(q^*)}(x-q^*) - \frac{7}{10}(x-q^*)^2$$

and

$$D(x) = \begin{cases} h(x), & 0\le x\le \tfrac12,\\ \ln 2, & \tfrac12 \le x \le z,\\ \frac{\ln 2}{h(z^2)}\,h(x^2), & z\le x\le 1, \end{cases}$$

and define the profile and the upper tail

$$\varphi(x) = \sqrt{\frac{D(x)h(x)}{h(q^*)}}\,e^{E(x)}, \quad s(x) = \sqrt{h(x^2)}. \tag{5}$$

With $\theta(x) = (x-A)/(B-A)$, the auxiliary function is $\psi = \varphi$ on $[0,A]$, $\psi = (1-\theta)\varphi + \theta s$ on $[A,B]$ and $\psi = s$ on $[B,1]$. The sampling coefficient is

$$a(x) = \begin{cases} 1, & 0\le x\le \tfrac12,\\ \sqrt{\dfrac{1-2x^2}{2x(1-x)}}, & \tfrac12 < x < \tfrac{1}{\sqrt2},\\ 0, & \tfrac{1}{\sqrt2}\le x\le 1. \end{cases} \tag{6}$$

Jiang takes the centre $x^* = 0.690908$, the scale $\sigma = 0.99999999$ and $mw_0 > \frac12$; here the centre is q^*, the scale is 1 and the weights are (4).[9] The middle formula of (6) is chosen so that

$$r(q,q) = q^2 + a(q)^2 q(1-q) = \tfrac12 \qquad \Big(\tfrac12 \le q \le \tfrac{1}{\sqrt2}\Big), \tag{7}$$

[7] For $0<q\le\frac12$ one has $f'(q) > q(1-2q) \ge 0$, since $-\ln(1-q) > q > \ln(1+q)$. On $[\frac12,1)$ the function f' is strictly decreasing, because $f''(q) = \ln\frac{1-q}{1+q} + \frac{1}{1+q}$ is negative at $\frac12$ and $f'''(q) = -\frac{2}{1-q^2} - \frac{1}{(1+q)^2} < 0$; it starts at $\frac12\ln\frac43 > 0$ and tends to $-\ln 2$. So f' has exactly one zero, which is q^*. At that zero $(1-q^*)\ln(1-q^*) = -q^*\ln(1+q^*)$, and expanding $h(q^2)$ gives $f(q^*) = \frac12(1-q^*)\ln(1+q^*)$; this is how (1) arises from $C^* = \frac12 - f(q^*)/\ln 2$.

[8] In integers: $2\cdot 181^2 = 65522 < 65536 = 256^2$, $2\cdot 179^2 = 64082 > 62500 = 250^2$ and $2\cdot 15^2 = 450 > 256$. These order relations, with $z^2+z>1$ and the position of the square $[0.682, 0.699]^2$ inside $[\frac12, z]^2$ and below A, are checked by the Lean kernel in the archive.

[9] The centre q^* is an irrational number given by (2). It is never approximated by a rational number in the proof: the interval computation carries an enclo-

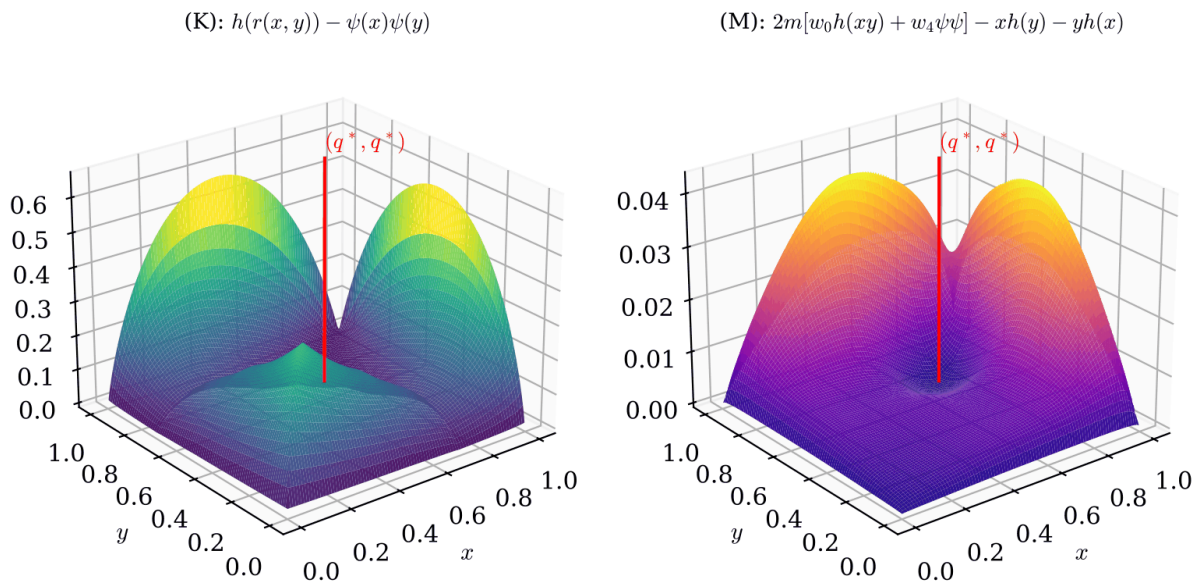


FIG. 2. The two gaps of the construction at the limit, over the unit square: on the left the gap $h(r(x,y)) - \psi(x)\psi(y)$ of the kernel inequality (K), on the right the gap $2m[w_0 h(xy) + w_4\psi(x)\psi(y)] - xh(y) - yh(x)$ of the mixture inequality (M), with $1-m = C^*$ and $mw_0 = \frac{1}{2}$. Both are nonnegative. Both vanish at the point (q^*, q^*) marked in red; (M) also vanishes on the boundary of the square, where both of its sides are zero, and (K) vanishes on the diagonal above $x = 0.721$, where $h(x^2) = s(x)^2$.

so the shared-randomness union bit has the largest possible entropy $\ln 2$ on that part of the diagonal.

IV. CONTACT AT ONE POINT AND CONVEXITY AROUND IT

Write

$$G_K(x,y) = K(x,y) - \psi(x)\psi(y),$$
$$G_M(x,y) = h(xy) + 2mw_4\psi(x)\psi(y) - xh(y) - yh(x),$$

so that (K) and (M) say $G_K \geq 0$ and $G_M \geq 0$ (for (M) we used $2mw_0 = 1$). Let $S = [q^* - 0.008, q^* + 0.008]^2$. On S every coordinate lies in $[0.682, 0.699] \subset (\frac{1}{2}, z)$ and below A, so $\psi = \varphi$, $D = \ln 2$ and a is given by its middle formula. On each of the two closed triangles of S cut by the diagonal, G_K agrees with one of the smooth functions

$$G_K^{(1)}(x,y) = h\big(xy + a(x)a(y)(x-xy)\big) - \varphi(x)\varphi(y),$$
$$G_K^{(2)}(x,y) = G_K^{(1)}(y,x),$$

the first on $x \leq y$ and the second on $x \geq y$.

Proposition IV.1. *The functions $G_K^{(1)}$, $G_K^{(2)}$ and G_M vanish at (q^*, q^*), and so do their gradients.*

Proof. By (7), $r(q^*, q^*) = \frac{1}{2}$, so $K(q^*, q^*) = \ln 2$. Since $D(q^*) = \ln 2$ and $E(q^*) = 0$, (5) gives $\varphi(q^*)^2 = \ln 2$; hence $G_K^{(i)}(q^*, q^*) = 0$. The logarithmic derivative of φ on $(\frac{1}{2}, z)$ is $\frac{h'}{2h} + E'$, and $E'(q^*) = -\frac{h'(q^*)}{2h(q^*)}$; so $\varphi'(q^*) = 0$. The gradient of $h(r)$ is $h'(r)\nabla r$, which vanishes at (q^*, q^*) because $h'(\frac{1}{2}) = 0$. The gradient of $\varphi(x)\varphi(y)$ is $(\varphi'(x)\varphi(y), \varphi(x)\varphi'(y))$, also zero there.

For G_M, the value at (q^*, q^*) is

$$h(q^{*2}) + 2mw_4 \ln 2 - 2q^* h(q^*) = 2mw_4 \ln 2 - 2f(q^*) = 0$$

by (4). Its partial derivative in x is

$$y h'(xy) + 2mw_4\varphi'(x)\varphi(y) - h(y) - y h'(x),$$

which at (q^*, q^*) equals $q^* h'(q^{*2}) - h(q^*) - q^* h'(q^*) = -f'(q^*) = 0$; by symmetry the partial derivative in y is also zero. □

At (q^*, q^*) the Hessians have closed forms (Appendix C). Put $\alpha = a(q^*)^2 = \frac{1-2q^{*2}}{2q^*(1-q^*)}$ and let $\kappa = (\ln\varphi)''(q^*)$. Then

$$\nabla^2 G_K^{(i)}(q^*, q^*) = -\alpha^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - \kappa \ln 2\, I, \tag{8}$$

$$\nabla^2 G_M(q^*, q^*) = \begin{pmatrix} \beta & \gamma \\ \gamma & \beta \end{pmatrix}, \tag{9}$$

where I is the identity matrix, $\beta = q^{*2} h''(q^{*2}) + 2f(q^*)\kappa - q^* h''(q^*)$ and $\gamma = h'(q^{*2}) + q^{*2} h''(q^{*2}) - 2h'(q^*)$. Numerically $\alpha = 0.10604564\ldots$, $\kappa = -6.03228523\ldots$, the matrix (8) has diagonal $4.17001582\ldots$ and off-diagonal $0.01124567\ldots$, and (9) has $\beta = 0.34256611\ldots$ and $\gamma = -0.21395816\ldots$. All

sure of q^* of radius 10^{-30} about 0.690907738725407754557261116505, obtained from a certified sign change of f', and every quantity that depends on q^* is evaluated over this interval. Since f' is strictly decreasing near q^*, the sign change proves that q^* lies in the interval.

TABLE I. Enclosures of the Hessians over the 1600 subsquares of S. The entries are the smallest lower bounds of g_{xx}, g_{yy} and of the determinant, and the largest bound of $|g_{xy}|$.

function	$\min g_{xx}$	$\min g_{yy}$	$\max \lvert g_{xy}\rvert$	$\min \det$
$G_K^{(1)}$	2.321	2.092	1.139	5.663
$G_K^{(2)}$	2.092	2.321	1.139	5.663
G_M	0.3114	0.3114	0.2303	0.0587

three are positive definite. Figure 3 shows the two gaps near the point of contact.

Proposition IV.2. *The Hessians of $G_K^{(1)}$, $G_K^{(2)}$ and G_M are positive definite at every point of S. Consequently $G_K \geq 0$ and $G_M \geq 0$ on S.*

Proof. Divide S into 40×40 congruent closed squares. On each of them we enclose the three second derivatives of each function by second-order forward differentiation in ball arithmetic with 200-bit precision, and check that the enclosures satisfy $f_{xx} > 0$, $f_{yy} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ for every choice of the entries in their balls. This implies positive definiteness at every point of the small square. Table I lists the extreme values over all 1600 squares.

A twice differentiable function g on a convex set with positive definite Hessian is convex. If moreover $g(p_0) = 0$ and $\nabla g(p_0) = 0$ at an interior point p_0, then $g(p) \geq g(p_0) + \nabla g(p_0)\cdot(p - p_0) = 0$ for every p in the set. Apply this to $G_K^{(1)}$, $G_K^{(2)}$ and G_M on S with $p_0 = (q^*, q^*)$, using Proposition IV.1. Since G_K equals $G_K^{(1)}$ or $G_K^{(2)}$ at each point of S, both gaps are nonnegative on S. □

The radius 0.008 is close to the largest that works for (K). The second derivatives of a grow without bound as $x \to 1/\sqrt{2} = 0.7071\ldots$, and at radius 0.01 the enclosures over squares near the upper corner of S are no longer positive definite.[10]

V. THE REST OF THE UNIT SQUARE

Away from S both gaps are strictly positive, and the covering of Jiang [9, Sections 6–9] applies with the new parameters. It splits the work into analytic estimates near the boundary of each domain and a box-by-box interval computation on the compact remainder. Figure 4 shows the layout. We list the parts and the scalar inequalities they rest on; Appendix D gives every estimate in full with the present parameters.

[10] This is a limitation of the enclosure and not of the inequality: G_K is positive at every point of $[0.682, 0.701]^2$ other than (q^*, q^*). The radius only has to be large enough for the box method of Section V to cover the complement of S, where both gaps are strictly positive.

TABLE II. The scalar inequalities used by the analytic parts, each checked by interval arithmetic with the parameters of Section III. Every margin is the lower end of a certified enclosure.

inequality	certified margin
$E'(1) > 0$	0.21766
$1 - C_g C_t > 0$	0.06286
$1 - C_b P_0 > 0$	0.51357
$1 - 2C_b P_0 > 0$	0.02714
$1 - E'(\frac{1}{2}) - h'(z^2)/h(z^2) > 0$	0.08172
$A - s(A)C_b > 0$	0.18250
$s(A) - \varphi(A) > 0$	0.00164
$-\big(\frac{h'(A)}{2h(A)} + E'(A)\big) > 0$	0.15960
$2mw_4(1 - \ln 0.004) - 1 > 0$	0.52752
$e^{-0.002} - B > 0$	0.27700
$16mw_4C^2 - \frac{1}{2} > 0$	0.13232

The kernel inequality. It suffices to prove $K \geq \varphi(x)\varphi(y)$ on the whole square, together with the bridge $h(Ay) \geq s(A)\varphi(y)$ for $0 \leq y \leq A$ and the inequality $\varphi \leq \psi \leq s$ on $[A, 1]$; the passage to ψ then uses the geometric concavity $h(xy) \geq s(x)s(y)$ (Lemma D.1). The strips $x \leq L$ and $x \geq U$ are handled by the constants

$$C_b = \frac{e^{E(L)}}{\sqrt{h(q^*)}}, \qquad P_0 = \frac{\ln 2\, e^{E(1/2)}}{\sqrt{h(q^*)}},$$

$$C_g = \sqrt{\frac{2\ln 2}{h(q^*)h(z^2)}}\, e^{E(1)}, \quad C_t = \frac{1}{U}\sqrt{\frac{\ln 2\, h(U^2)h(U)}{h(z^2)h(q^*)}}\, e^{E(1)},$$

through the scalar inequalities of Table II. The compact square $[L, U]^2$ is covered by boxes in Jiang's parametrisation of the middle interval by $t = a(x)$, which removes the singularity of a' at $1/\sqrt{2}$. Boxes contained in S are accepted by Proposition IV.2.

The mixture inequality. With $x = e^{-u}$, $y = e^{-v}$, $u = t(1 + \xi)$, $v = t(1 - \xi)$, $G(u) = e^u h(e^{-u})$ and $P(u) = e^u \psi(e^{-u})$, inequality (M) on the half square $x \leq y$ is equivalent to

$$F(t, \xi) = \tfrac{1}{2}G(2t) + mw_4 P(u)P(v) - \tfrac{1}{2}\big(G(u) + G(v)\big) \;\geq\; 0 \tag{10}$$

for $t > 0$, $0 \leq \xi \leq 1$. For $t \leq 0.001$ the bound $G(u) + G(v) - G(u + v) \leq 2\sqrt{uv}$ and $P = \sqrt{G(2\cdot)}$ give $F \geq \big[2mw_4(1 - \ln 0.004) - 1\big]\sqrt{uv} \geq 0$. For $t \geq 16$ the lower bound $\psi \geq Ch$ with $C = e^{E(0)}/\sqrt{h(q^*)}$ gives $F \geq (16mw_4C^2 - \frac{1}{2})G(v) \geq 0$. Near the edge $\xi \to 1$, where $v \to 0$, two estimates give $F \geq (mw_0 - \frac{1}{2})G(2t)$, which is 0 by (4); this suffices, because the acceptance test only needs $F \geq 0$. The remaining rectangle $[0.001, 16] \times [0, 1]$ is covered by interval boxes, and boxes whose image lies in S are accepted by Proposition IV.2.

The computation. Every quantity is an interval with outward rounding at 35 significant digits. A box is accepted when the interval value of the gap has a positive lower end, or when the value at its centre minus the derivative bound times the radius is positive, or when an analytic edge bound or Proposition IV.2 applies; otherwise it is bisected. A completed run accepts every descendant of the initial boxes, which proves coverage. Table III records the run; it takes about ninety sec-

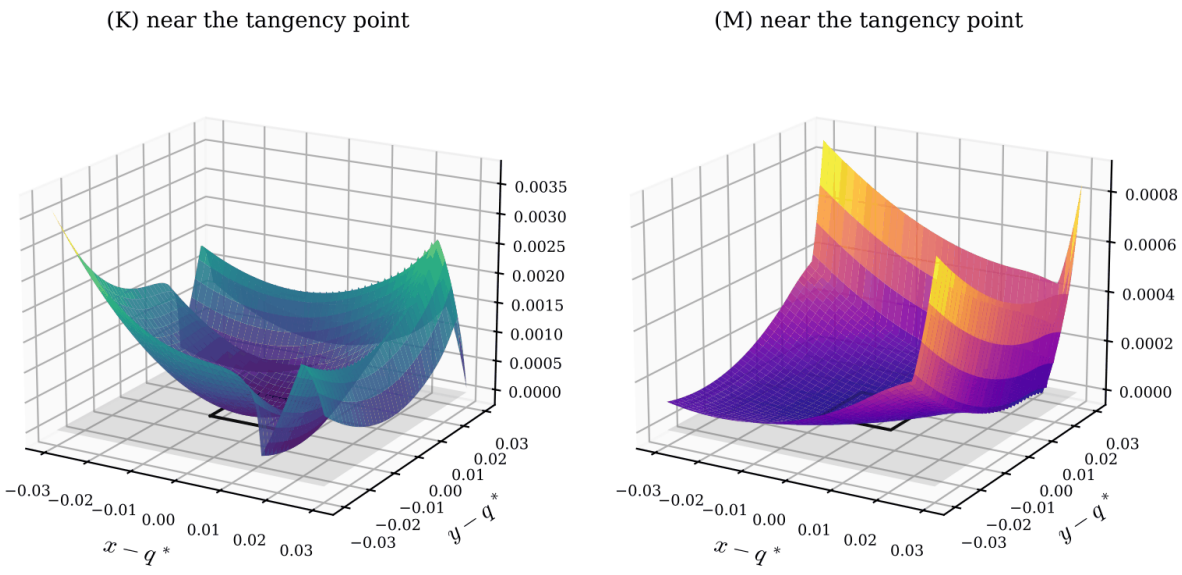


FIG. 3. The gaps of (K) and (M) on $[q^* - 0.03, q^* + 0.03]^2$, plotted against $(x - q^*, y - q^*)$. Each surface touches the grey plane of height zero only at the origin. The black square is the boundary of $S = [q^* - 0.008, q^* + 0.008]^2$, on which the Hessians are certified positive definite. The ridge of the left surface along the diagonal is the kink of $\min(x, y)$ in (3).

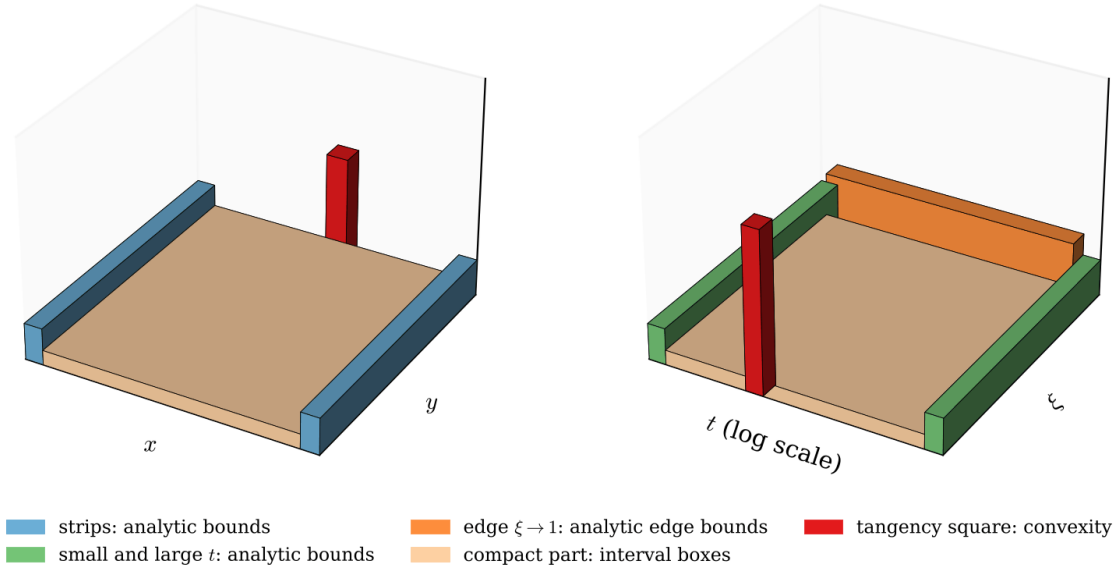


FIG. 4. How the two domains are covered. Left: the unit square of (K), with the lower and upper strips $x \le \frac{1}{16}$ and $x \ge \frac{15}{16}$ handled analytically, the compact square handled by interval boxes in Jiang's three coordinate maps, and the square S around (q^*, q^*) handled by convexity. Right: the domain of (M) in the coordinates $x = e^{-t(1+\xi)}$, $y = e^{-t(1-\xi)}$, with analytic bounds for $t < 0.001$, for $t \ge 16$ and near the edge $\xi \to 1$, interval boxes elsewhere, and the image of S handled by convexity.

onds on one processor.

VI. PROOF OF THE THEOREMS

Proof of Theorem A. By Proposition IV.2, (K) and (M) hold on S. On the rest of the unit square they hold by Section V and

TABLE III. The interval verification at the limit. “Convex” counts the boxes accepted because they lie in S; “edge” counts the mixture boxes accepted by the edge estimates. The smallest margin is over the boxes accepted with a positive lower bound.

part	boxes	leaves	convex	edge	smallest margin
bridge	39	20	–	–	4.60×10^{-6}
kernel	1572	789	53	–	1.44×10^{-7}
mixture	8193	4101	23	91	7.10×10^{-8}

Appendix D: (K) by the strips, the compact square, the bridge and the upper tail, and (M) on $x \le y$ by the small and large t estimates, the edge estimates and the compact rectangle, and on $x > y$ by symmetry, while both sides of (M) vanish when $xy = 0$ or $x = y = 1$. Proposition II.1 with $1 - m = C^*$ gives $p \ge C^*$. □

Proof of Theorem B. Appendix F repeats Jiang’s argument [9, Proposition 10.3] with the present parameters. It gives $p \ge 1 - m + mw_0 V/(16\ln^2 N)$, where $V \ge \ln^2(2N-1)/(4N^2)$ is a variance of logarithms of union multiplicities. With $1 - m = C^*$ and $mw_0 = \frac{1}{2}$ this is the stated bound. □

VII. BEYOND THE LIMIT

Theorem A closes the gap between what the reduction (K)–(M) can certify and what has been certified with it. Further progress on Frankl’s conjecture through entropy needs a different inequality. Three facts constrain the search.[11] First, the edge $y = 1$ of (M) and the diagonal bound $\psi^2 \le \ln 2$ alone force $1 - m \le C^*$, whatever a and ψ are, and the same holds for any finite sum of products $\psi_j(x)\psi_j(y)$. Second, any single-letter certificate that uses the independent protocol and other protocols admitting component hiding is bounded by C^*. Third, any single-letter certificate whose classes contain product laws is bounded by $1 - h(1/\sqrt{2})/\sqrt{2} = 0.383099\ldots$.

A useful structural fact holds for every union-closed family: if the next bit of a prefix α is forced to be zero, so is the next bit of every prefix $\alpha' \subseteq \alpha$ that occurs.[12] So the set of prefixes at which an element is surely absent is a down-set. The adversarial laws behind the ceilings place such prefixes freely. An argument that uses the down-set structure, for instance through a correlation inequality of Harris type across coordinates, is not covered by the ceilings; we do not know whether it gives more.

VIII. FAMILIES OF AT MOST FIFTY-TWO MEMBERS

In this section a *counterexample* is a finite union-closed family $\mathscr{A}$ with a nonempty member in which every element lies in fewer than half of the members. Fix a counterexample $\mathscr{A}$ with the least possible number of members, write $U = \bigcup \mathscr{A}$ (a member) and $m = |U|$. For $i \in U$ let A_i be the union of the members avoiding i, which is the largest member avoiding i, and write $|x|$ for the number of members containing x. We say that j *dominates* x, and write $x \preceq j$, if every member containing x contains j; then $x \notin A_j$ exactly when $x \preceq j$, so $D(j) = U \setminus A_j$ is the set of elements dominated by j.[13]

Lemma VIII.1. *$\mathscr{A}$ may be taken separating, $|\mathscr{A}| = 2n+1$ with every frequency at most n and some frequency equal to n, $m \ge 13$ and $n \ge 2m-1$.*

The first three claims are standard; the fourth is the theorem for at most 12 elements [16], and the last is the bound of Roberts and Simpson [13], in the form proved by Hu [8]. Appendix G proves all but the fourth. An element of frequency n is called a *top*.

The boundary case. Suppose now $n = 2m-1$, so $|\mathscr{A}| = 4m-1$. Appendix G shows that every top t has a *partner* p, another top, such that the members containing both t and p are exactly U and the A_i with $i \in W = U \setminus \{t, p\}$. Counting then splits $\mathscr{A}$ into four blocks by membership of t and p (Figure 5): P_{11} with $m-1$ members and P_{10}, P_{01}, P_{00} with m members each.[14] For a member X of P_{10} put $R_X = W \setminus X$, and for $Y \in P_{01}$ put $S_Y = W \setminus Y$. Let $\mathscr{R}$ and $\mathscr{S}$ be the two resulting families of subsets of W, and for $x \in W$ let $r(x)$ and $s(x)$ count the members of $\mathscr{R}$ and $\mathscr{S}$ that contain x.

Lemma VIII.2. *Write $\le$ for domination restricted to W. Then:*

(a) $\le$ is a partial order on W, and $D(j) = \{x \in W : x \le j\}$ for $j \in W$;

[11] We record them because they decide where effort is useful. None of them is new: the first is Jiang’s Proposition 4.1 [9], the second and third are the two ceilings of Moffat [11]. The search for routes past C^* was run with the BLLM engine Anaxia [20], which holds the known constructions for a problem, each typed by what it gives and what it needs, breeds candidates by combining them, and accepts a candidate only when a program checks it. For this paper the engine supplied the status check: its indexes of DeepMind’s formal-conjectures collection and of the OpenRefute archive list every recorded variant of the union-closed conjecture with its status. It also recorded each route tried beyond C^* and why it failed. One of them is a conjecture of Roos [14]: a union-closed family whose largest frequency is exactly one half has 2^k members. It holds for all 2,771,104 union-closed families on at most five elements, but it implies Frankl’s conjecture (a smallest counterexample $\mathscr{F}$ has odd size and gives the family $\{A, A\cup\{z\} : A \in \mathscr{F}\}$ on one new element z, which has frequency exactly one half and $2|\mathscr{F}|$ members), so it gives no shortcut. The construction of Section III, the tangency and convexity proofs and the interval certificate were made and verified by the programs of the archive [21], not by the engine.

[12] If $\alpha' \subseteq \alpha$ are both prefixes of members and some member extending α' had bit 1 at the next coordinate, its union with a member extending α would extend α and have bit 1 there.

[13] Domination is the order of the join-irreducible elements of the lattice $\mathscr{A}$ read backwards. It is reflexive and transitive, and on a separating family it is antisymmetric, because two elements dominating each other lie in the same members.

[14] The block P_{10} consists of the members containing t and not p, and so on. The sizes follow from $|t| = |p| = 2m-1$ and $|P_{11}| = m-1$: $|P_{10}| = |P_{01}| = (2m-1)-(m-1) = m$ and $|P_{00}| = (4m-1)-(m-1)-2m = m$.

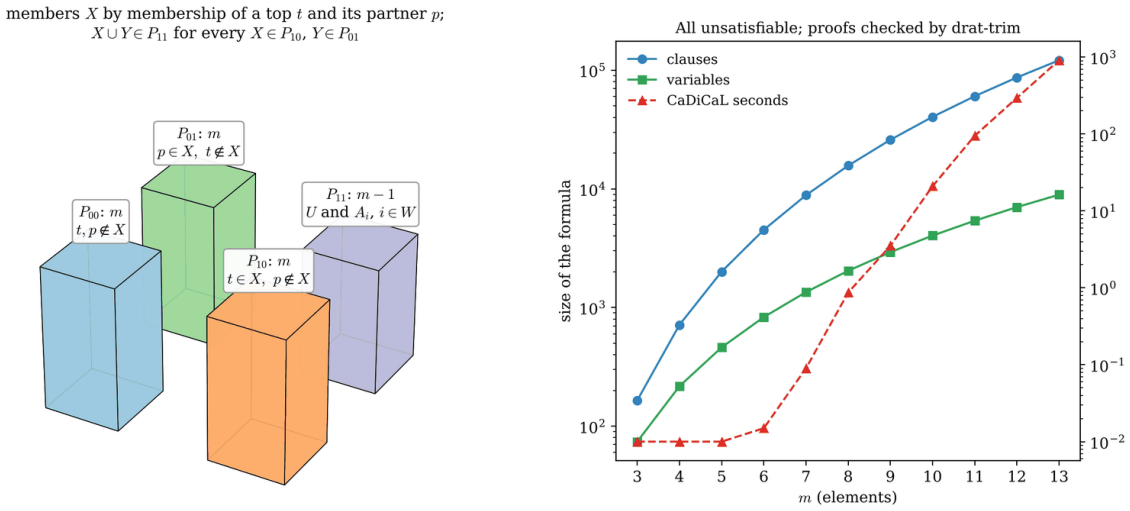


FIG. 5. The boundary case $|\mathscr{A}| = 4m-1$. Left: the four blocks of members by membership of a top t and its partner p; every union of a member of P_{10} and one of P_{01} falls into the $m-1$ members of P_{11}. Right: the size of the satisfiability problem of Theorem VIII.4 and the time CaDiCaL takes to refute it, for $m = 3, \ldots, 13$.

(b) $\mathscr{R}$ and $\mathscr{S}$ each consist of m distinct down-sets of $(W, \leq)$ and are closed under intersection;

(c) for every $R \in \mathscr{R}$ and $S \in \mathscr{S}$, $R \cap S$ is empty or equal to $D(j)$ for some $j \in W$;

(d) $r(x) + s(x) \geq m-1$ for every $x \in W$.

The proof is in Appendix G. Conditions (a) to (d) mention only W, which has $m-2$ points, and two families of m sets; they no longer describe a union-closed family, and the order is unknown. Nevertheless they are contradictory, by hand when the order is trivial and by computer for $m \leq 13$.

Proposition VIII.3. *For every $m \geq 4$ there is no configuration as in Lemma VIII.2 in which $\leq$ is trivial. Hence in a smallest counterexample with $4m-1$ members, for every top t and partner p, some element of $U \setminus \{t, p\}$ is dominated by another.*

The proof, a counting argument on the sizes of the largest members of $\mathscr{R}$ and $\mathscr{S}$, is in Appendix G. When $\leq$ is not trivial, $R \cap S$ may be a down-set with several points and the count loses its slack; we have not found a hand argument for that case.

Theorem VIII.4. *For $3 \leq m \leq 13$ there is no configuration as in Lemma VIII.2, whatever the partial order.*

Proof. Encode (a) to (d) in propositional logic, with one variable for each pair $x < j$ of a naturally labelled order on $W = \{0, \ldots, m-3\}$ (every partial order has a linear extension, so the labels may be chosen to increase along the order), one for each membership $x \in R$ and $x \in S$, and selector variables for the intersections required by (b) and (c). Transitivity, the down-set property and the selectors are clauses of at most four literals; distinctness is imposed by listing the members of each family in strictly increasing lexicographic order, which loses no generality; (d) is a cardinality constraint written as a sequential counter. For $m = 13$ the formula has 8,929 variables and 121,290 clauses. The solver CaDiCaL reports each formula unsatisfiable and writes a DRAT proof, and the independent checker drat-trim verifies every proof.[15] □

Proof of Theorem C. Suppose a counterexample with at most 52 members exists, and let $\mathscr{A}$ be one with the fewest members. By Lemma VIII.1, $|\mathscr{A}| = 2n+1 \leq 52$, so $n \leq 25$, while $n \geq 2m-1 \geq 25$. Hence $m = 13$ and $n = 2m-1$, which is the boundary case. Lemma VIII.2 gives a configuration on 11 points with $m = 13$, and Theorem VIII.4 says that none exists. □

Remark VIII.5. The next sizes are not reached by the same method. A smallest counterexample with 53 members has $n = 26$ and, by Lemma VIII.1, $m = 13$, so $n = 2m$ lies one step above the boundary case and the blocks no longer have the sizes used above. The configuration of Lemma VIII.2 for $m = 14$ (a counterexample with 55 members on 14 elements) is a formula of the same kind, about four times larger; it also needs the theorem for 13-element ground sets, which is open.

[15] The formula is written by `checks/sat/boundary_cnf.py` and the solver and checker are run by `checks/sat/run_sat.sh` in the archive [21]. For $m = 13$ the solver needs about 15 minutes on one core and its proof has about one gigabyte. Two further checks guard the encoding. Weakening (d) to $r + s \geq m-3$ makes the formula satisfiable for $m = 7, 8$, and the decoded solutions satisfy (a) to (d) in that weakened form when tested directly, so the clauses do not exclude more than they should. A second encoding, written separately on the python-sat library, gives the same answer for every $m \leq 13$ (with its own proof for $m = 13$, also verified by drat-trim), and for $m = 6, 7, 8$ the result agrees with integer programs solved separately for each nontrivial partial order on 4, 5 and 6 points (15, 62 and 317 orders up to isomorphism).

DATA AND CODE AVAILABILITY

The interval programs, the Hessian certification, the encoder of Theorem VIII.4 with the script that runs CaDiCaL and checks its proofs with drat-trim, independent checks in C, C++ and Julia, the Lean 4 check of the rational breakpoints, the figure programs and the LaTeX source are in the repository `github.com/creelie/eUUR` and its Zenodo archive, doi:10.5281/zenodo.23480218 [21]. The literature and status checks and the record of the routes tried beyond C^* were made with the BLLM engine Anaxia, release v3.0.5 [20]. Deep Bhattacharjee carried out the computations and the Python, C, C++, Julia, Lean and shell verifications with AI assistance (Claude, Anthropic).

ETHICAL DECLARATIONS

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AUTHOR CONTRIBUTIONS

Deep Bhattacharjee proposed the problem, carried out the computations and wrote the paper. Ushashi Bhattacharya checked the derivations of the appendices. Both authors approved the final version.

Appendix A: Proof of the entropy reduction

This appendix proves Proposition II.1, following Gilmer [7], Liu [10] and Jiang [9]. We use two elementary facts about h: it is concave, so $h(\lambda x) \geq \lambda h(x)$ for $\lambda \in [0,1]$, and conditioning reduces entropy.

Native conditional probabilities. Let $\mathscr{F} \subseteq 2^{[n]}$ with $N \geq 2$ and let X be uniform on $\mathscr{F}$, viewed as a 0–1 vector. For a prefix $\alpha \in \{0,1\}^{i-1}$ set $x_i(\alpha) = \Pr(X_i = 0 \mid X_{<i} = \alpha)$, and let P_i be the law of $x_i(X_{<i})$. Then

$$\mathbb{E}_{P_i} x = 1 - p_i, \qquad \mathbb{E}_{P_i} h(x) = H(X_i \mid X_{<i}), \tag{A1}$$

where p_i is the frequency of element i.

The independent rule. Take two independent copies $X^{(0)}$ and $Y^{(0)}$ of X, and let $Z_0 = X^{(0)} \vee Y^{(0)}$, the coordinatewise maximum, which is the indicator of the union. Given both prefixes at coordinate i, the two bits are independent with zero probabilities $S = x_i(X^{(0)}_{<i})$ and $T = x_i(Y^{(0)}_{<i})$, so the union bit is zero with probability ST. The prefix of Z_0 is a function of the two prefixes, so

$$H\big((Z_0)_i \mid (Z_0)_{<i}\big) \;\geq\; H\big((Z_0)_i \mid X^{(0)}_{<i}, Y^{(0)}_{<i}\big) = \mathbb{E}_{P_i \otimes P_i} h(ST). \tag{A2}$$

The shared rule. At each coordinate i draw a fresh uniform U_i on $[0,1]$, shared by the two samples, and set $Q_U(x) = x + a(x)(\mathbf{1}\{U \leq x\} - x)$. Given U_i and the two prefixes, generate the two bits independently with zero probabilities $Q_{U_i}(S)$ and $Q_{U_i}(T)$, where S, T are the native probabilities of the two prefixes. Since $\mathbb{E}_U Q_U(x) = x$ and U_i is independent of each prefix, each sample has its native transition law at every step, so by induction each of $X^{(4)}, Y^{(4)}$ is uniform on $\mathscr{F}$. Since $\mathbb{E}\,\mathbf{1}\{U \leq x\}\mathbf{1}\{U \leq y\} = \min(x,y)$,

$$\mathbb{E}_U\big[Q_U(x) Q_U(y)\big] = xy + a(x)a(y)\big(\min(x,y) - xy\big) = r(x,y),$$

so given the prefixes the union bit of $Z_4 = X^{(4)} \vee Y^{(4)}$ is zero with probability $r(S,T)$ and has entropy $K(S,T)$.

Conditional independence. The shared variables used before coordinate i form the vector $V_i = (U_1, \ldots, U_{i-1})$. Given V_i, the two prefixes are functions of V_i and of independent local randomness, so they are independent and identically distributed; hence so are S and T. Therefore

$$\mathbb{E}[\psi(S)\psi(T)] = \mathbb{E}\Big[\big(\mathbb{E}[\psi(S) \mid V_i]\big)^2\Big] \geq \big(\mathbb{E}\psi(S)\big)^2 = \big(\mathbb{E}_{P_i}\psi\big)^2,$$

by the nonnegativity of a variance. With (K) and conditioning,

$$H\big((Z_4)_i \mid (Z_4)_{<i}\big) \;\geq\; \mathbb{E}K(S,T) \;\geq\; \big(\mathbb{E}_{P_i}\psi\big)^2. \tag{A3}$$

Combination. Integrate (M) against $P_i \otimes P_i$. The right side becomes $2(\mathbb{E}_{P_i} x)(\mathbb{E}_{P_i} h) = 2(1-p_i)H(X_i \mid X_{<i})$ by (A1), and the product term on the left becomes $(\mathbb{E}_{P_i}\psi)^2$ because the two variables are independent. Dividing by $2m$,

$$w_0\, \mathbb{E}_{P_i \otimes P_i} h(ST) + w_4 \big(\mathbb{E}_{P_i}\psi\big)^2 \;\geq\; \frac{1-p_i}{m} H(X_i \mid X_{<i}).$$

By (A2) and (A3), summing over i and using the chain rule,

$$w_0 H(Z_0) + w_4 H(Z_4) \;\geq\; \sum_{i=1}^{n} \frac{1-p_i}{m} H(X_i \mid X_{<i}) \;\geq\; \frac{1-p}{m} \ln N,$$

since $1 - p_i \geq 1 - p$ and $\sum_i H(X_i \mid X_{<i}) = H(X) = \ln N$. Union closure puts Z_0 and Z_4 in $\mathscr{F}$, so each has entropy at most $\ln N$ and the left side is at most $\ln N$. Dividing by $\ln N > 0$ gives $p \geq 1 - m$. If $N = 1$ the only member is nonempty and some element has frequency 1.

Appendix B: Why the limit is C^*

Suppose (K) and (M) hold. Since $r(1,1) = 1$, (K) gives $\psi(1)^2 \leq h(1) = 0$, so $\psi(1) = 0$. Put $y = 1$ in (M): the left side is $2mw_0 h(x)$ and the right side is $h(x)$, so

$$mw_0 \;\geq\; \tfrac{1}{2}. \tag{B1}$$

Put $x = y = q$ in (M) and use $\psi(q)^2 \leq K(q,q) \leq \ln 2$ and $w_4 = 1 - w_0$:

$$\begin{aligned} 2q\,h(q) &\leq 2mw_0 h(q^2) + 2(m - mw_0)\ln 2 \\ &= 2m\ln 2 - 2mw_0\big(\ln 2 - h(q^2)\big) \\ &\leq 2m\ln 2 - \ln 2 + h(q^2), \end{aligned}$$

using (B1) and $h(q^2) \leq \ln 2$. Hence $1 - m \leq \frac{1}{2} - f(q)/\ln 2$ for every q, and the best choice $q = q^*$ gives $1 - m \leq C^*$.

Equality $1 - m = C^*$ therefore requires $mw_0 = \frac{1}{2}$, $\psi(q^*)^2 = K(q^*,q^*) = \ln 2$, and equality in (M) at (q^*,q^*). These are exactly the conditions imposed in Section III. Since G_K and G_M are nonnegative and vanish at the interior point (q^*,q^*), their gradients must vanish there too, so Proposition IV.1 is forced as well; what has to be shown is the second-order condition.

Appendix C: The Hessians at the point of contact

We derive (8) and (9). Throughout, $q = q^*$, $h'(u) = \ln\frac{1-u}{u}$ and $h''(u) = -\frac{1}{u(1-u)}$.

The profile. On $(\frac12, z)$ we have $\ln\varphi = \frac12\ln\ln 2 + \frac12\ln h - \frac12\ln h(q) + E$, so

$$(\ln\varphi)' = \frac{h'}{2h} + E', \qquad (\ln\varphi)'' = \frac{h''h - h'^2}{2h^2} - \frac{7}{5}.$$

At q the first derivative is zero. Since $\varphi''/\varphi = (\ln\varphi)'' + ((\ln\varphi)')^2$, we get $\varphi''(q)\varphi(q) = \kappa\,\varphi(q)^2 = \kappa\ln 2$ with

$$\kappa = \frac{h''(q)h(q) - h'(q)^2}{2h(q)^2} - \frac{7}{5} = -6.0322852376\ldots$$

The gradient of r. On the branch $\min(x,y) = x$ we have $r = xy + a(x)a(y)(x-xy)$, so at (q,q), writing $a = a(q)$ and $a' = a'(q)$,

$$r_x = q + aa'q(1-q) + a^2(1-q), \quad r_y = q + aa'q(1-q) - a^2 q.$$

Differentiating the identity $a(x)^2 x(1-x) = \frac12 - x^2$ at q gives $2aa'q(1-q) + a^2(1-2q) = -2q$, that is $aa'q(1-q) = -q - \frac12\alpha(1-2q)$ with $\alpha = a^2$. Substituting,

$$r_x = \tfrac12\alpha, \qquad r_y = -\tfrac12\alpha.$$

On the other branch the roles are exchanged and $\nabla r = (-\frac12\alpha, \frac12\alpha)$.

The kernel gap. The Hessian of $h(r)$ is $h''(r)\nabla r\nabla r^{\mathsf T} + h'(r)\nabla^2 r$. At (q,q), $r = \frac12$, $h'(\frac12) = 0$ and $h''(\frac12) = -4$, so the Hessian of $h(r)$ is $-4\nabla r\nabla r^{\mathsf T} = -\alpha^2\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ on either branch. The Hessian of $\varphi(x)\varphi(y)$ at (q,q) is $\left(\begin{smallmatrix} \varphi''\varphi & \varphi'^2 \\ \varphi'^2 & \varphi\varphi'' \end{smallmatrix}\right) = \kappa\ln 2\, I$. Subtracting gives (8). Its eigenvalues are $-\kappa\ln 2 = 4.1812\ldots$ on the vector $(1,1)$ and $-2\alpha^2 - \kappa\ln 2 = 4.1588\ldots$ on $(1,-1)$, both positive.

The mixture gap. On S the mixture gap is $G_M = h(xy) + 2mw_4\varphi(x)\varphi(y) - xh(y) - yh(x)$. Differentiating twice,

$$\begin{aligned}\partial_{xx}G_M &= y^2 h''(xy) + 2mw_4\varphi''(x)\varphi(y) - yh''(x),\\ \partial_{xy}G_M &= h'(xy) + xyh''(xy) + 2mw_4\varphi'(x)\varphi'(y)\\ &\qquad - h'(y) - h'(x).\end{aligned}$$

At (q,q), with $\varphi'(q) = 0$, $\varphi''(q)\varphi(q) = \kappa\ln 2$ and $mw_4\ln 2 = f(q)$, these become β and γ of (9). Numerically $\beta = 0.3425661126\ldots$ and $\gamma = -0.2139581685\ldots$, so the eigenvalues $\beta+\gamma = 0.1286\ldots$ and $\beta-\gamma = 0.5565\ldots$ are positive.

From the point to the square. Positive definiteness at (q,q) gives nonnegativity only in some neighbourhood of unknown size. Proposition IV.2 makes the neighbourhood explicit. Second-order forward differentiation propagates triples (value, gradient, Hessian) through sums, products, quotients, exp, ln and $\sqrt{\cdot}$ by the chain rule; over a box each entry is a ball containing all values the true entry takes there. Because the entries are bounded independently, the test $g_{xx} > 0$, $g_{yy} > 0$, $g_{xx}g_{yy} - g_{xy}^2 > 0$ on the balls proves positive definiteness at every point of the box.

Appendix D: The analytic estimates

This appendix gives the analytic parts of the covering with the parameters of Section III. The arguments are those of Jiang [9, Sections 6–8]; we repeat them so that the paper is self-contained, and because three of them use $mw_0 - \frac12$, which is positive in Jiang's setting and zero here. The margins quoted are those of Table II.

Lemma D.1 (geometric concavity)**.** *For $0 < x, y < 1$, $h(xy) \ge s(x)s(y)$. For fixed $0 < y \le A$, the ratio $h(xy)/s(x)$ is nondecreasing on $[A, 1)$.*

Proof. Let $\rho(x) = xh'(x)/h(x)$ and $\ell(x) = -\ln(1-x)$, so that $h = xh' + \ell$. Then $h^2\rho' = h'\ell - \frac{h}{1-x}$. For $x \ge \frac12$ the first term is at most 0 and the second is negative. For $x < \frac12$, substituting $h = xh' + \ell$ gives $h^2\rho' = h'(\ell - \frac{x}{1-x}) - \frac{\ell}{1-x} < 0$, because $h' > 0$ and $\ell(x) = \int_0^x \frac{dt}{1-t} < \frac{x}{1-x}$. So ρ decreases. For $g(t) = \ln h(e^t)$, $t < 0$, we have $g'(t) = \rho(e^t)$ and $g''(t) = e^t\rho'(e^t) < 0$; concavity of g at $2\ln x$ and $2\ln y$ gives $\ln h(xy) \ge \frac12\ln h(x^2) + \frac12\ln h(y^2)$. For the second claim, $\frac{d}{dx}\ln\frac{h(xy)}{s(x)} = \frac{\rho(xy)-\rho(x^2)}{x} \ge 0$ because $xy \le x^2$. □

The upper tail. Since $A > z$, every $x \ge A$ satisfies

$$\frac{\varphi(x)}{s(x)} = \sqrt{\frac{\ln 2\,h(x)}{h(z^2)h(q^*)}}\, e^{E(x)}.$$

Its logarithmic derivative $\frac{h'}{2h} + E'$ has derivative $\frac{h''h-h'^2}{2h^2} - \frac75 < 0$, so it decreases; at A it is negative (margin 0.15960) and $s(A) > \varphi(A)$ (margin 0.00164). So φ/s starts below 1 and decreases, which gives $\varphi \le \psi \le s$ on $[A, 1]$, and $\psi \ge \varphi$ everywhere.

From φ to ψ. Suppose $K \ge \varphi(x)\varphi(y)$ on the square and $h(Ay) \ge s(A)\varphi(y)$ on $[0, A]$. If $x, y \le A$ then $\psi = \varphi$. If $x, y \ge A$ then $a(x) = a(y) = 0$ and $K = h(xy) \ge s(x)s(y) \ge \psi(x)\psi(y)$. If $x \ge A \ge y$, then $K = h(xy) \ge s(x)h(Ay)/s(A) \ge s(x)\varphi(y) \ge \psi(x)\psi(y)$ by Lemma D.1.

Elementary bounds on φ. Since $E'' = -\frac75$ and $E'(1) > 0$ (margin 0.21766), E is increasing on $[0, 1]$. For $x \le L$, $D = h$ and $\varphi \le C_b h(x)$. For $x \le \frac12$, $\varphi \le P_0$. For $\frac12 < x < 1$ the logarithmic derivative of $\varphi(x)/x$ is $\frac{D'}{2D} + \frac{h'}{2h} + E' - \frac1x \le \frac{h'(z^2)}{h(z^2)} + E'(\frac12) - 1 < 0$ (margin 0.08172), so $\varphi(y) \le 2P_0 y$ on $[\frac12, 1]$.

The lower strip. Let $x \le L$ and $x \le y$. If $y \le \frac12$ then $r = x$ and $\varphi(x)\varphi(y) \le C_b P_0 h(x) \le h(x) = K$ (margin 0.51357). If $y \ge \frac12$ then $r = xk$ with $k = y + a(y)(1-y) \in [y, 1]$, and concavity gives $K = h(xk) \ge kh(x) \ge yh(x) \ge 2C_b P_0\, y\, h(x) \ge \varphi(x)\varphi(y)$ (margin 0.02714).

The upper strip. For $x \ge U$ the four bounds $h(x) \le h(U)$, $h(x^2) \le h(U^2)$, $\frac1x \le \frac1U$ and $E(x) \le E(1)$ give $\varphi(x) \le C_t x$. For all y, $D(y)/h(y) \le 2\ln 2/h(z^2)$ (using $h(z^2) \le 2h(z)$), so $\varphi(y) \le C_g h(y)$. Since $a(x) = 0$, $K = h(xy) \ge xh(y) \ge C_t C_g\, xh(y) \ge \varphi(x)\varphi(y)$ (margin 0.06286).

The bridge. If $y \le L$, then $\varphi(y) \le C_b h(y)$ and $h(Ay) \ge Ah(y)$, so

$$h(Ay) - s(A)\varphi(y) \ge \big(A - s(A)C_b\big)h(y) \ge 0$$

(margin 0.18250). For $L \le y \le A$ the interval computation gives $h(Ay) - s(A)\varphi(y) \ge 4.60 \times 10^{-6}$.

Facts about G. With $G(u) = u - (e^u - 1)\ln(1 - e^{-u})$ one has $G(u) = u + 1 - \sum_{j \ge 1} \frac{e^{-ju}}{j(j+1)}$, hence $G' > 0$, $G'' < 0$ and $G(u) \ge u$. The function $J(u) = G(u) + u \ln u$ is convex with $J(0) = 0$, which gives $G(u) \ge u(1 - \ln u)$ and $G(u) + G(v) - G(u+v) \le (u+v)h(\frac{u}{u+v}) \le 2\sqrt{uv}$.

Small t. If $t \le T = 0.001$, then $x, y \ge e^{-0.002} > B$ (margin 0.27700), so $\psi = s$ and $P(u) = \sqrt{G(2u)}$. Hence

$$\begin{aligned} F &\ge (mw_0 - \tfrac{1}{2})G(2t) + mw_4\sqrt{G(2u)G(2v)} - \sqrt{uv} \\ &\ge \big[2mw_4(1 - \ln 4T) - 1\big]\sqrt{uv} \;\ge\; 0, \end{aligned}$$

using $mw_0 = \frac{1}{2}$ and $\sqrt{G(2u)G(2v)} \ge 2\sqrt{uv}(1 - \ln 4T)$ (margin 0.52752).

Large t. One has $D \ge h$ on $[0,1]$ (for $x \ge z$ this uses $z^2 + z > 1$, so that $h(x^2) \ge h(x)$), hence $\psi \ge \varphi \ge Ch$ and $P \ge CG$. For $t \ge 16$, $G(u) \ge u \ge 16$, $G(2t) \ge G(u)$ and

$$\begin{aligned} F &\ge \tfrac{1}{2}G(u) - \tfrac{1}{2}G(u) + \big(mw_4C^2G(u) - \tfrac{1}{2}\big)G(v) \\ &\ge \big(16mw_4C^2 - \tfrac{1}{2}\big)G(v) \;\ge\; 0 \end{aligned}$$

(margin 0.13232). Here $mw_0 = \frac{1}{2}$ is used in the first step.

The edge $\xi \to 1$. Since $G(u) \le G(2t)$ and the product term is nonnegative, $F \ge (mw_0 - \frac{1}{2})G(2t) - \frac{1}{2}G(v)$. If moreover $v \le -\ln B$ and $G(v) \le [2mw_4CG(t)]^2$, then $P(v) = \sqrt{G(2v)} \ge \sqrt{G(v)}$ and $P(u) \ge CG(t)$, so $mw_4P(u)P(v) \ge \frac{1}{2}G(v)$ and $F \ge (mw_0 - \frac{1}{2})G(2t) = 0$. By (4) this bound is zero, and a box on which it applies is accepted with lower bound 0. This suffices, since (M) only asks for $F \ge 0$.

Appendix E: The interval method

Each number is an interval $[\ell, u]$ with ℓ rounded down and u rounded up at 35 significant decimal digits; logarithms, exponentials and square roots are enlarged by one unit in the last place on each side. The binary entropy on an interval is bounded by its endpoint values and, if the interval contains $\frac{1}{2}$, by $\ln 2$. For a box $I \times J$ with centre (c,d) and radii ρ_I, ρ_J, if $|g_x| \le M_x$ and $|g_y| \le M_y$ on the box, then

$$g(x,y) \;\ge\; g(c,d) - M_x\rho_I - M_y\rho_J.$$

At the joins of piecewise definitions both one-sided derivatives are included. A box is accepted if the interval value of the gap has positive lower end, or if the bound above is positive, or (for the mixture) if an edge estimate of Appendix D applies, or if the box lies in S (for the kernel, its image under the coordinate map; for the mixture, its image under $(t,\xi) \mapsto (e^{-t(1+\xi)}, e^{-t(1-\xi)})$). The test for S uses the inner enclosure $[q^*_{\mathrm{hi}} - 0.008,\; q^*_{\mathrm{lo}} + 0.008]$, which lies inside S for every value of q^* in its enclosure. A box whose centre has a negative upper bound would be a counterexample; none occurs. Otherwise the box is split in two along the coordinate with the larger error term. The kernel starts from the six unordered pairs of Jiang's three coordinate maps on $[L, \frac{1}{2}]$, $[\frac{1}{2}, \frac{1}{\sqrt{2}}]$ and $[\frac{1}{\sqrt{2}}, U]$; the mixture starts from nine rectangles with t-knots $0.001, 0.01, 0.1, 0.3, 0.5, 1, 2, 4, 8, 16$. The run of Table III accepts every box.

Appendix F: The strict inequality

Let $N \ge 2$ and let $r(S)$ be the number of ordered pairs $(A,B) \in \mathscr{F}^2$ with $A \cup B = S$. Jiang [9, Section 10] constructs a coupling of two uniform members, by a relative-entropy projection of the inverse-multiplicity weighting onto the couplings with uniform marginals, whose union Z_* satisfies

$$H(Z_*) \;\ge\; H(Z_0) + \frac{V}{16\ln N}, \qquad V \;\ge\; \frac{\ln^2(2N-1)}{4N^2},$$

where Z_0 is the union of two independent uniform members and V is the double-centred variance of $\ln r$ [9, Theorem 10.2]. We now run Appendix A with Z_* in place of Z_0. Both Z_* and Z_4 are unions of two members of $\mathscr{F}$, so they take values in $\mathscr{F}$ and

$$\ln N \;\ge\; w_0H(Z_*) + w_4H(Z_4).$$

The bound on $H(Z_*)$ above and the combination step of Appendix A then give

$$\begin{aligned} w_0H(Z_*) + w_4H(Z_4) &\ge w_0H(Z_0) + w_4H(Z_4) + \frac{w_0V}{16\ln N} \\ &\ge \frac{1-p}{m}\ln N + \frac{w_0V}{16\ln N}. \end{aligned}$$

Multiplying by $m/\ln N$ gives $p \ge 1 - m + mw_0V/(16\ln^2 N)$. With $1 - m = C^*$ and $mw_0 = \frac{1}{2}$,

$$p \;\ge\; C^* + \frac{V}{32\ln^2 N} \;\ge\; C^* + \frac{\ln^2(2N-1)}{128N^2\ln^2 N}.$$

Appendix G: The boundary configuration

This appendix proves Lemma VIII.1 (except $m \ge 13$), the existence of partners, Lemma VIII.2 and Proposition VIII.3. Throughout, $\mathscr{A}$ is a counterexample with the fewest members.

Separation. If two elements x, y lie in the same members, deleting y from every member gives a union-closed family with the same number of members (two members cannot differ only in y) and the same frequencies. So among the counterexamples with the fewest members we may take one on the fewest elements, and it is separating: any two elements are distinguished by a member.

Odd size. A member B that is minimal under inclusion is not a union of other members, so $\mathscr{A} \setminus \{B\}$ is again union-closed, and it still has a nonempty member. If $|\mathscr{A}| = 2k$, every frequency is at most $k - 1 < \frac{2k-1}{2}$, so $\mathscr{A} \setminus \{B\}$ would be a smaller counterexample. Hence $|\mathscr{A}| = 2n+1$ and every frequency is at most n. If every frequency were at most $n-1$, removing B would again give a counterexample. So some element has frequency n.

Tops lie in every A_i. Let t be a top and $i \ne t$. If $t \notin A_i$, then $t \preceq i$, so the members containing t are among those containing i; since $|i| \le n = |t|$ they are the same members, against separation. So $t \in A_i$. The members A_i are pairwise distinct:

if $A_i = A_j$ with $i \neq j$, a member containing exactly one of i, j, say j, avoids i and so lies in A_i, which then contains j. Hence $\mathscr{S}_t = \{U\} \cup \{A_i : i \neq t\}$ is a set of m distinct members containing t.

The bound $n \geq 2m-1$. The family $\mathscr{A}^t$ of members avoiding t has $n+1$ members, is union-closed and has A_t as its largest member. If $n = 0$ then $\mathscr{A} = \{U\}$, which is not a counterexample, so $n \geq 1$ and $A_t \neq \emptyset$. By minimality $\mathscr{A}^t$ is not a counterexample, so the set Q of elements lying in at least $\frac{n+1}{2}$ members of $\mathscr{A}^t$ is not empty. Choose $p \in Q$ maximal for $\preceq$. If $i \notin \{t, p\}$ and $p \notin A_i$, then $p \preceq i$, every member of $\mathscr{A}^t$ containing p contains i, so $i \in Q$, and $p \preceq i$ contradicts the maximality of p. So p lies in U and in every A_i with $i \notin \{t, p\}$, that is, in $m-1$ members of $\mathscr{S}_t$, and

$$n \;\geq\; |p| \;\geq\; \frac{n+1}{2} + m - 1,$$

which is $n \geq 2m-1$.[16]

Partners. If $n = 2m-1$, every inequality in the display is an equality. So p is a top, it lies in exactly m members of $\mathscr{A}^t$ and in exactly $m-1$ members containing t, and these are U and the A_i with $i \in W = U \setminus \{t, p\}$. This p is a partner of t.

Proof of Lemma VIII.2. (a) Domination is reflexive and transitive, and antisymmetric by separation. For $j \in W$ we have $D(j) = U \setminus A_j$, and $t, p \in A_j$ by the two paragraphs above, so $D(j) \subseteq W$.

(b) A member $X \in P_{10}$ is $\{t\} \cup (W \setminus R_X)$, so $X \mapsto R_X$ is injective and $\mathscr{R}$ has m members. If $j \in R_X$ and $x \leq j$, then $x \notin X$, because $x \in X$ would force $j \in X$; so R_X is a down-set. For $X, X' \in P_{10}$ the union $X \cup X'$ lies in P_{10} and $R_{X \cup X'} = R_X \cap R_{X'}$. The same holds for $\mathscr{S}$.

(c) For $X \in P_{10}$ and $Y \in P_{01}$, the union $X \cup Y$ contains t and p, so it is U or some A_j with $j \in W$. Hence $R_X \cap S_Y = W \setminus (X \cup Y)$ is empty or $W \setminus A_j = D(j)$.

(d) Let $x \in W$. At least $(4m-1) - (2m-1) = 2m$ members avoid x: $r(x)$ of them in P_{10}, $s(x)$ in P_{01}, at most m in P_{00}, and in P_{11} the members A_j with $x \in D(j)$, whose number is $d(x) = |\{j \in W : x \leq j\}|$. So $r(x) + s(x) \geq m - d(x)$. Take $j \in W$ maximal with $x \leq j$. Then $d(j) = 1$, so $r(j) + s(j) \geq m-1$, and since the members of $\mathscr{R}$ and $\mathscr{S}$ are down-sets, $r(x) \geq r(j)$ and $s(x) \geq s(j)$. □

Proof of Proposition VIII.3. Suppose $\leq$ is trivial, so $D(j) = \{j\}$ and, by (c), $|R \cap S| \leq 1$ for all $R \in \mathscr{R}$, $S \in \mathscr{S}$. Write $w = m-2$. By (b) each family has a least member, R_0 and S_0, contained in all its members; $|R_0 \cap S_0| \leq 1$.

Case $R_0 \cap S_0 = \{z\}$. Then $R \cap S = \{z\}$ for all R, S. A point $y \neq z$ lying in some R and some S would give $|R \cap S| \geq 2$, so every $y \neq z$ lies only in members of $\mathscr{R}$ or only in members of $\mathscr{S}$; call these sets of points X_R and X_S. By (d), a point of X_R has $r \geq m-1$, so it is missing from at most one member of $\mathscr{R}$. The sets $X_R \setminus R$, $R \in \mathscr{R}$, are therefore pairwise disjoint, and they are distinct because the members $R = \{z\} \cup (R \cap X_R)$ are distinct. At most one of them is empty, so $|X_R| \geq m-1$; likewise $|X_S| \geq m-1$. But $|X_R| + |X_S| \leq w - 1 = m-3$.

Case $R_0 \cap S_0 = \emptyset$. Let α, β be the largest sizes in $\mathscr{R}$ and $\mathscr{S}$, and let $\delta_R = \sum(\alpha - |R|)$ over the $m-1$ members $R \neq R_0$, and δ_S likewise. A point of R_0 lies in all m members of $\mathscr{R}$ and a point of S_0 in all of $\mathscr{S}$. Counting incidences with (d),

$$\begin{aligned} |R_0| + |S_0| + (m-1)(\alpha+\beta) - \delta_R - \delta_S &= \sum_{x \in W} \big(r(x) + s(x)\big) \\ &\geq (m-1)\, w + |R_0| + |S_0|, \end{aligned}$$

so $\delta_R + \delta_S \leq (m-1)(\alpha + \beta - w)$. A largest R and a largest S meet in at most one point, so $\alpha + \beta \leq w+1$, and the inequality forces $\alpha + \beta \geq w$.

If $\alpha + \beta = w$, then $\delta_R = 0$: every $R \neq R_0$ has size α, and two of them meet in a smaller member, which can only be R_0. So the $m-1$ sets $R \setminus R_0$ are nonempty and pairwise disjoint in $w - |R_0| < m-1$ points, which is impossible.

So $\alpha + \beta = w+1$ and $\delta_R + \delta_S \leq m-1$. Now a largest R and a largest S meet in exactly one point and together cover W. Let k_R, k_S be the numbers of largest members. Each member other than the least and the largest ones adds at least 1 to δ, so $k_R + k_S \geq 2(m-1) - (\delta_R + \delta_S) \geq m-1 \geq 3$, and we may assume $k_R \geq 2$. If $R_1 \neq R_2$ are largest, every largest S contains $(W \setminus R_1) \cup (W \setminus R_2) = W \setminus (R_1 \cap R_2)$, which has at least $w - \alpha + 1 = \beta$ points; so $|R_1 \cap R_2| = \alpha - 1$ and $S = W \setminus (R_1 \cap R_2)$. Thus $k_S = 1$, the largest member O of $\mathscr{S}$ is unique, and every two largest members of $\mathscr{R}$ meet in $T = W \setminus O$, so they are $T \cup \{x\}$ for distinct points $x \in O$, where $|O| = \beta = w + 1 - \alpha$. Since $k_S = 1$, $\delta_S \geq m-2$, hence $\delta_R \leq 1$. The set $T = R_1 \cap R_2$ belongs to $\mathscr{R}$ and contains R_0.

If $T = R_0$, every $R \neq R_0$ strictly contains T and so has size at least α: all $m-1$ of them are largest and give $m-1$ distinct points of O, while $|O| = w + 1 - \alpha \leq m-2$. If $T \neq R_0$, then T has size $\alpha - 1$ and adds 1 to δ_R, so $\delta_R = 1$ and the other $m-2$ members besides R_0 and T are largest; they give $m-2$ distinct points of O, while $|O| = w + 1 - \alpha \leq m-3$ because $\alpha \geq |T| + 1 \geq |R_0| + 2 \geq 2$. Both are impossible. □

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